

Question	Answer	Mark	Mark scheme	Additional guidance
21	984.(3677853) and 969.(0181643)	B1	stating bound of 51.95 or 52.05 or 1.445 or 1.455	Accept 52.049 or 52.0499... for 52.05 Accept 1.4549 or 1.4549... for 1.455
		P1	for process to rearrange formula to give g as the subject, eg $g = \frac{4\pi^2 l}{T^2}$ oe	Rearrangement may occur after substitution, in this case correct bounds are not needed for this mark
		P1	for process to use LB of l and UB of T in formula for g or T or process to use UB of l and LB of T in formula for g or T eg $\frac{4\pi^2 [\text{LB of } l]}{[\text{UB of } T]^2}$ or $\frac{4\pi^2 [\text{UB of } l]}{[\text{LB of } T]^2}$	$51.95 \leq [\text{LB of } l] < 52.0$ $1.45 < [\text{UB of } T] \leq 1.455$ $52.0 < [\text{UB of } l] \leq 52.05$ $1.445 \leq [\text{LB of } T] < 1.45$ Rearrangement may not be correct
		A1	for upper bound = 984.(3677853) or 984.(1125639..) and lower bound = 969.(0181643) or 968.(7669227..)	NB: correct answer without supportive working gets 0 marks Accept answers rounded or truncated to 3sf or better